MobilityhubTools: applications to the Australian context

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Introduction

Shared mobility hubs are an emerging approach in transport systems, aiming to improve sustainability and many other aspects of transport and landuse systemsˆ["...a place where different transport modes are integrated seamlessly, promoting efficient and sustainable urban mobility. Emphasis is given to shared mobility options such as bikes, scooters and cars. Connection with public transport (transit) is desirable but not necessary. The integration between mobility suppliers is important to provide a seamless, flexible connection at these transfer points. Ensuring an enjoyable experience for travelers is a crucial part of the hub concept as well, and therefore the hub should be considered not just as a transfer node but as a place where a variety of services come together, such as logistics services or restaurants" (Roukouni et al. 2023).].[1] Recent research has explored shared mobility hub information provision and started the development of tools for practitioners, tactical urbanists and others who might wish to implement their own hub.[2] However, as of yet these tools have not been used on hubs (or potential hub locations) outside of Ireland and the UK. It is unclear whether this approach, based on Open Street Maps datasets, will work elsewhere, while there is also a need for cross-case comparison.

This research note explores the use of the mobilityHubTools R package on sites within Australia. It seeks to improve understanding of the generalisability of the mobilityHubTool software, and whether this approach is worth persuing further. The note details the application of the software to a small number of cases in Australia, and reports findings about the performance of the tools and directions for future research.

[1] Anastasia Roukouni et al., "An Analysis of the Emerging 'Shared Mobility Hub' Concept in European Cities: Definition and a Proposed Typology," Article, *Sustainability (Switzerland)* 15, no. 6 (2023), doi:10.3390/su15065222.

[2] James Reynolds et al., "Do-It-Yourself Information with mobilityHubTools," in *Collaborative European Research Conference (CERC)*, 2026.

Background

Various guidance has already been developed for the provision of user informational displays at shared mobility hubs.For example. CoMoUK,[3] CoMoUK[4] and CoMoUK[5] provide some insights as to preferred practices. However, most mobility hub informational displays, maps and associated signage appears to be bespoke, and largely delivered on a site-by-site or region-by-region basis. For example,

[3] *Communications: How to Increase Awareness and Support for Mobility Hubs*, 2021, `https://tinyurl.com/5cxuxeru`.

[4] *Mobility Hub Delivery Models; Funding, Procurement and Management Guidance*, 2021.

[5] *Mobility Hub Toolkits*, 2021, `https://tinyurl.com/ux2svctn`.

Figure 1 provides an example of branding and guidance at a hub in Bremen, Germany (via CoMoUK[6]), while Figures 2 and 3 show images from the authors' visits to Birmhimgham and Bristol undertaken as part of the research described in Reynolds et al.[7]

[6] *Communications*.

[7] "Do-It-Yourself Information with mobilityHubTools."

Branding

A typical hub with Bremen's branding: a reduced carriageway and user-friendly setting showing totem pole branded mobil.punkt (in English 'mobility point')

Bremen has developed a clear brand recognised by users and non-users alike. Hubs are located where they are highly visible and with a clearly recognised totem pole to increase awareness.

Public awareness is promoted by adverts featuring Udo– a character whose name stands for 'Use it don't own it'.

Udo shows the positive impact of shared mobility on lifestyle. Udo can be seen on large roadside hoardings, street posters, adverts on public transport and at interchanges, in cinema video adverts and leaflets.

This approach has won over the public to see the advantages of mobility hubs and some streets are even asking the council to install a hub in their street.

Looking at further developing these ideas, the city of Cologne offers car-free days in neighbourhoods with the street and parking space used for sustainability related stalls, shared transport operators vehicles on show, games, busking, local shops and cafés opening longer hours. These are promoted as trying the 'good life' rather than banning cars. By creating a fun and relaxed atmosphere, people can see what reduced traffic and parking could feel like.

Figure 1: CoMoUK increasing awareness case study. Source: CoMoUK (2021)

A BRIEF SCAN suggests that there has been much mobility-hub-related activity in the Australasian context, including:

- research about site selection in Melbourne;[8]
- trials of four mobility hubs at Melbourne Airport, seeking to improve access and navigation around car parks, pick-up and drop-off areas, check-in and baggage reclaim[9];
- planning for a mobility hub in Meadowbrook, within the City of Logan (Queensland)[10]; and
- construction of a mobility hub at Glen Eden Train Station in Auckland[11].

[8] Allan Pimenta et al., "Site Selection for Future Mobility Hubs in Melbourne: A Multicriteria Location-Allocation Analysis," *Findings*, 2024, doi:10.32866/001c.94389.

[9] `https://www.instagram.com/p/DBsn0WsoLS5/`

[10] `https://www.haveyoursaylogan.com.au/projects/project-11-meadowbrook-mobility-hub`

[11] `https://tinyurl.com/bdzsupe5`

Figure 2: Images of the mobility hub at Andrew Road, Birmingham. Source: authors.

Figure 3: , Images of mobility hubs in Bristol. Source: authors.

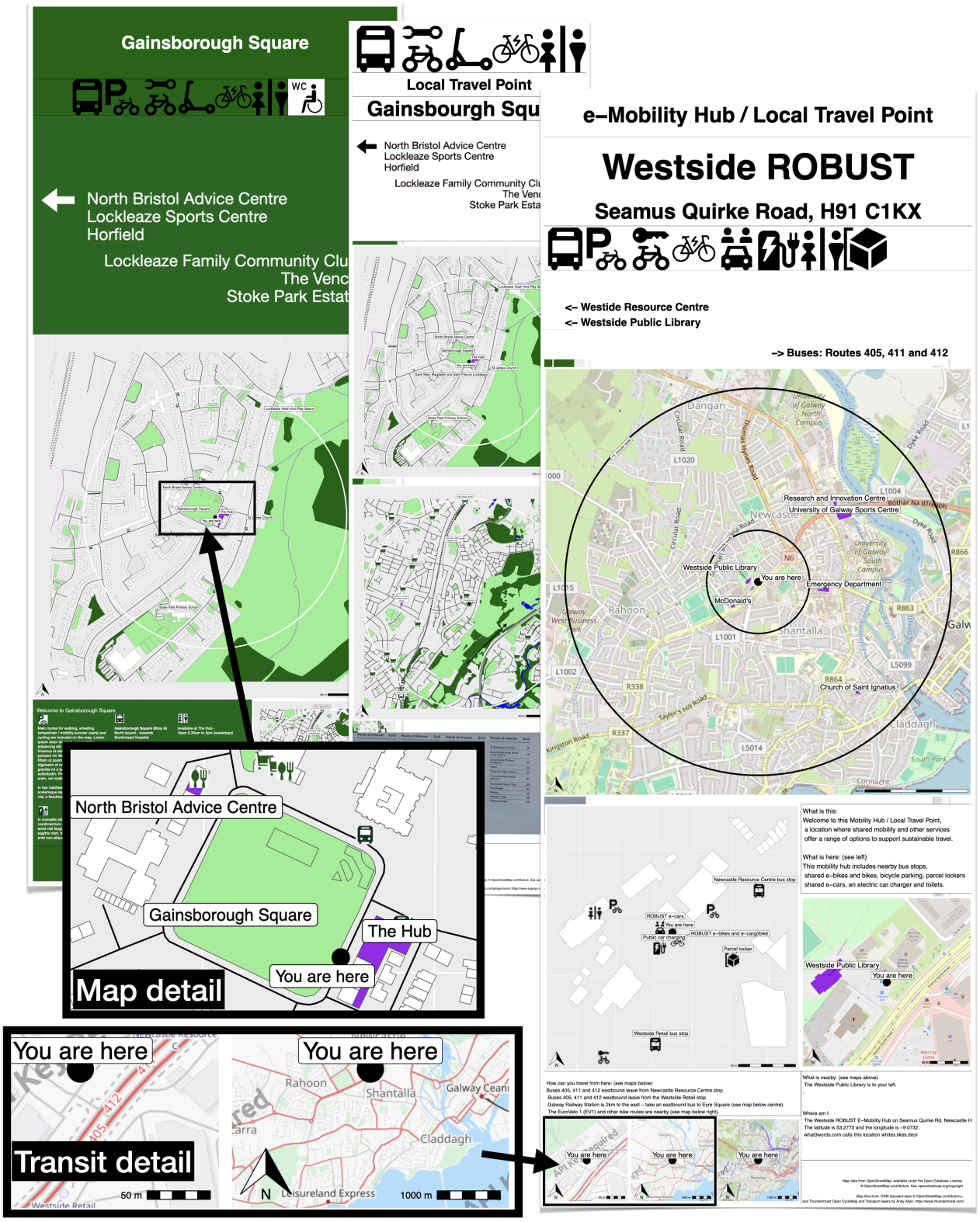

Figure 4: Software output examples: Bristol, Birmingham and synthesis-style totem displays (left, centre, right). Source: authors.

Methodology

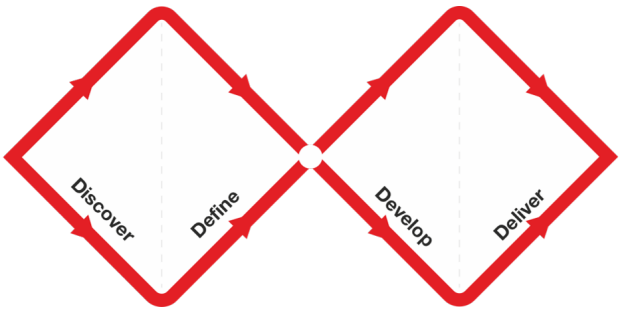

Figure 5: Framework for Innovation. Source : UK Design Council (undated)

Results

- *Discovery:* wide range of examples[12].

[12] https://maps.app.goo.gl/1dd8sEJCiwCgPkc98, https://maps.app.goo.gl/j7b7MLr8HzcjqoS66, https://maps.app.goo.gl/BW8cEVXioYSoNbiz9, https://maps.app.goo.gl/1JqvPLWkwkbnyGB27

Discussion and conclusions

- Tactical urbanism: why not make a "mobility hub" of your own?
- More to do to improve the code, develop further outputs etc. (Open source, so you can help with this).
- Open Street Maps data quite powerful, as are the GIS tools available for R.